

# Thermal reliability selects a detector cost, not a universal action

A receiver with a fixed activated-rate prefactor cannot be made arbitrarily ready while remaining quiet for a prescribed duration and responding quickly to a signal. In the model below, those two requirements impose an exact lower bound on signal-induced barrier reduction. Interpreting that reduction as incident energy needs an additional coupling premise. The bound retains bath temperature, timing and error tolerances; it does not select a quantum action.

This is Q10's exploratory effective-model test, with a written consistency review and no accepted-ledger promotion. The exponential escape law and its signal coupling are physical hypotheses. They are not a derivation of escape from Q09's conservative double well or a microscopic model of photon detection.

## 1. A receiver with a specified retention requirement

Let  $T > 0$  be bath temperature,  $k_B$  Boltzmann's constant, and  $\nu > 0$  an attempt frequency in inverse seconds. A prepared metastable receiver has residual activation barrier  $b \geq 0$ , measured in energy units. Its first recorded escape is an absorbing event. The stipulated dark rate is

$$r_0 = \nu e^{-b/(k_B T)}. \quad (1)$$

Thus the survival probability  $S$  solves  $dS/dt = -r_0 S$ ,  $S(0) = 1$ , giving  $S(t) = \exp(-r_0 t)$ . Absorption describes a monitored first event; maintaining a permanent physical record and resetting it remain separate apparatus tasks. Retention here means absence of a false first event while armed.

During a signal gate of duration  $t_g > 0$  the barrier is reduced by an amount  $u \geq 0$ , held constant throughout the gate. Use the explicitly capped model

$$r_1 = \nu e^{-\max(b-u, 0)/(k_B T)}. \quad (2)$$

Here  $u$  denotes an energy change, whereas  $\nu$  denotes a rate. The cap prevents an unphysical extrapolation to rates above the stipulated attempt frequency. It is a model definition, not a Kramers formula near a vanishing barrier. The same  $\nu$  in (1) and (2), an unchanged bath and exponential holding times are essential premises. Sustaining  $u$  may require work from a control source; no equality between  $u$  and input pulse energy has yet been assumed.

Require false-event probability at most  $\epsilon$  over a dark interval  $\tau > 0$ , and miss probability at most  $\eta$  in a subsequent signal gate, conditional on surviving to that gate. For  $0 < \epsilon, \eta < 1$ , define

$$a = -\log(1 - \epsilon) > 0, \quad c = \log(1/\eta) > 0. \quad (3)$$

The two requirements are exactly

$$r_0 \tau \leq a, \quad r_1 t_g \geq c. \quad (4)$$

Conditional efficiency matters: the probability of both surviving the dark interval and firing in the gate is  $\exp(-r_0 \tau)[1 - \exp(-r_1 t_g)]$ . The separate bounds ensure it is at least  $(1 - \epsilon)(1 - \eta)$ .

## 2. Exact feasibility and the optimal residual barrier

Timely detection is impossible in this model if  $\nu t_g < c$ , even with zero remaining barrier. If  $\nu t_g \geq c$ , define

$$B = \max\left(0, k_B T \log \frac{\nu \tau}{a}\right), \quad D = k_B T \log \frac{\nu t_g}{c} \geq 0. \quad (5)$$

The feasible designs are precisely

$$b \geq B, \quad u \geq \max(0, b - D). \quad (6)$$

Indeed, taking logarithms of (4) first gives  $b \geq k_B T \log(\nu\tau/a)$ . The signal inequality gives  $\max(b - u, 0) \leq D$ . Since  $D \geq 0$ , this is equivalent to  $b - u \leq D$ , proving (6). Every point in (6) satisfies both original probability bounds, including the equality cases. Minimizing over  $b \geq 0$  therefore yields

$$u_{\min} = \max(0, B - D) = k_B T \left[ \log \frac{c\tau}{at_g} \right]_+, \quad \nu t_g \geq c, \quad (7)$$

where  $[x]_+ = \max(x, 0)$ . For the last equality, if  $\nu\tau \leq a$  then  $B = 0$ ; feasibility also gives  $c\tau/(at_g) \leq \nu\tau/a \leq 1$ . If  $\nu\tau > a$ , subtract the two logarithms in (5). This checks both branches without extending a negative barrier.

In the discriminating regime  $c\tau > at_g$ , choose  $b = B$  and  $u = B - D$ . Both probability constraints then saturate. If  $c\tau \leq at_g$ , a design with  $u = 0$  can already satisfy them: the timing/error specification is too weak to require signal discrimination. A nonzero temperature alone is insufficient.

### 3. What becomes of preloading and a slow receiver?

At fixed  $T, \nu, \tau, \epsilon$  with  $\nu\tau > a$ , preloading cannot reduce  $b$  below  $B$  while satisfying dark retention. This closes the zero-residual-barrier route within the specified metastable class. A receiver with bare barrier  $\Delta$  and a sustainable preload  $e_0$ , modeled by  $b = \Delta - e_0$ , must satisfy

$$0 \leq e_0 \leq \Delta - B. \quad (8)$$

If  $\Delta < B$  the design cannot meet retention. A particle initially at rest near Q09's unstable saddle is not automatically such a sustainable metastable preload: it moves even without noise. Equation (8) applies only if an actual preparation/control mechanism realizes (1) for the full dark interval.

Reducing  $\nu$  alone improves dark survival but slows detection. The gate floor  $\nu \geq c/t_g$  prevents using  $\nu \rightarrow 0$  at fixed gate and miss tolerance. In the positive-bound regime the same prefactor cancels from (7), exposing the required contrast  $r_1/r_0 \geq c\tau/(at_g)$ . Retention alone would not give this protection: it is the combined dark and signal requirement that does so.

If the signal can instead change the prefactor, a rate-only alternative is

$$r_0 = \nu_0 e^{-b/(k_B T)}, \quad r_1 = \nu_1 e^{-b/(k_B T)}. \quad (9)$$

For any fixed  $b \geq 0$  choose  $\nu_0 = (a/\tau) \exp(b/(k_B T))$  and  $\nu_1 = (c/t_g) \exp(b/(k_B T))$ . Both specifications hold with  $u = 0$ . This is an exact counterexample to an inference from probabilities alone to barrier reduction, not a construction of a zero-work autonomous detector. Changing mobility, coupling or access to a reaction channel has its own physical control cost, which (9) does not specify.

### 4. Energy, action and the limits of the bound

To infer incident work  $W$ , add the explicit transduction premise

$$u \leq \chi W, \quad 0 < \chi < \infty, \quad (10)$$

with fixed dimensionless  $\chi$  and with all sources contributing to barrier control included in  $W$ . Then (7) implies

$$W \geq \frac{k_B T}{\chi} \left[ \log \frac{c\tau}{at_g} \right]_+ . \quad (11)$$

A stored-energy amplifier need not obey (10) for signal work alone. Nor is a barrier-height change automatically dissipated heat or reset work: it concerns the energy landscape along an escape path, not the work along the actual control trajectory. Q09’s cycle accounting remains necessary.

For Q08’s narrow-band wave input of angular frequency  $\omega$ , canonical wave action is approximately  $W/\omega$ . Under (10) its corresponding lower bound is the right side of (11) divided by omega. A finite-pulse wave-action definition must retain Q08’s bandwidth qualification. Alternatively  $Wt_g$  has action units but describes a different quantity. Neither choice removes T, error rates, timing, transduction or frequency from the result.

Within the stipulated rate-model class,  $T \rightarrow 0$  at fixed  $\nu$  and fixed specifications sends the optimal b, u and W bound to zero while preserving all rate ratios. No microscopic low-temperature extrapolation is claimed. Allowing larger  $\chi$  also removes a bound on signal work; allowing  $t_g \geq c\tau/a$  removes the positive discrimination requirement. These are explicit class limits, not quantum predictions.

## 5. Strategic consequence and evidence map

Thermal reliability supplies a conditional cost for a ready detector once a response deadline and a fixed signal coupling are included. It does not by itself identify a universal action or yield exclusive two-receiver events. Park further Arrhenius/threshold variants. The next useful construction is an autonomous shared-resource receiver: can a finite mechanical latch make two classical outputs compete for one stored excitation while retaining reliable response? Specify its resource, locality and output probabilities. That tests Q09’s exclusivity obstruction directly and can fail by requiring communication or a prepared shared resource; it must not assume a Born rule.

Source context is limited. The existing B23 companion, Zwanzig p. 219 (21)–(24), supplies a bath-memory construction, not (1). One discovery search located Hänggi, Talkner and Borkovec, *Reaction-rate theory: fifty years after Kramers*, Rev. Mod. Phys. 62, 251–341 (1990), DOI. Its bibliographic metadata/abstract identify activated escape as established context. The PDF was opened but no rate formula is imported from it; this is not a passage or prior-art audit. Equations (1), (2) and (10) are declared premises; (3)–(8) and (11) are written consequences; (9) tests an omitted coupling premise. Independent proof and bounded literature review remain required before ledger promotion. No computational verification was used.